# Class Field Theory I.3.6

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## Original statement

*Source CFT.pdf, physical page 47.*

THEOREM 3.6 *Let $K_{\pi,n}$ denote the subfield $K[\Lambda_n]$ of $K^{\mathrm{al}}$ generated over $K$ by the elements of $\Lambda_n$.*

*(a) For each $n$, $K_{\pi,n}/K$ is totally ramified of degree $(q-1)q^{n-1}$.*

*(b) The action of $A$ on $\Lambda_n$ defines an isomorphism*

$$(A/\mathfrak{m}^n)^\times \longrightarrow \mathrm{Gal}(K_{\pi,n}/K).$$

*In particular, $K_{\pi,n}/K$ is abelian.*

*(c) For each $n$, $\pi$ is a norm from $K_{\pi,n}$.*

## Original proof

*Source CFT.pdf, physical pages 47–48.*

PROOF. Again, we may assume that $f \in \mathcal{F}_\pi$ is a polynomial of the form $\pi T + \cdots + T^q$.

(a), (b). Choose a nonzero root $\pi_1$ of $f(T)$ and (inductively) a root $\pi_n$ of $f(T) - \pi_{n-1}$. Consider the sequence of fields

$$K[\Lambda_n] \supset K[\pi_n] \overset{q}{\supset} K[\pi_{n-1}] \overset{q}{\supset} \cdots \overset{q}{\supset} K[\pi_1] \overset{q-1}{\supset} K.$$

Each extension is Eisenstein (ANT, 7.55) with the degree indicated. Therefore $K[\pi_n]$ is totally ramified over $K$ of degree $q^{n-1}(q-1)$.

Recall that $\Lambda_n$ is the set of roots of $f^{(n)}$ in $K^{\mathrm{al}}$, and so $K[\Lambda_n]$ is the splitting field of $f^{(n)}$. Therefore $\mathrm{Gal}(K[\Lambda_n]/K)$ can be identified with a subgroup of the group of permutations of the set $\Lambda_n$, but Lemma 3.5 implies that each element of $\mathrm{Gal}(K[\Lambda_n]/K)$ acts on $\Lambda_n$ as an $A$-module isomorphism, and so the image of $\mathrm{Gal}(K[\Lambda_n]/K)$ in the automorphism group of $\Lambda_n$ is contained in

$$\mathrm{End}_A(\Lambda_n) = (A/(\pi^n))^\times.$$

Hence

$$(q-1)q^{n-1} \geq |\mathrm{Gal}(K[\Lambda_n]/K)| = [K[\Lambda_n] : K] \geq [K[\pi_n] : K] = (q-1)q^{n-1}.$$

We must have equalities throughout, and so $\mathrm{Gal}(K[\Lambda_n]/K) \simeq (A/\mathfrak{m}^n)^\times$ and $K[\Lambda_n] = K[\pi_n]$.

(c) Let $f^{[n]}(T) = (f/T) \circ f \circ \cdots \circ f$ ($n$ terms), so that

$$f^{[n]}(T) = \pi + \cdots + T^{(q-1)q^{n-1}}.$$

Then $f^{[n]}(\pi_n) = f^{[n-1]}(\pi_{n-1}) = \cdots = f(\pi_1) = 0$. Because $f^{[n]}$ is monic of degree $(q-1)q^{n-1} = [K[\pi_n] : K]$, it must be the minimal polynomial of $\pi_n$ over $K$. Therefore,

$$\begin{aligned} \mathrm{Nm}_{K[\Lambda_n]/K}\, \pi_n &= (-1)^{(q-1)q^{n-1}}\pi \\ &= \pi \quad \text{unless } q = 2 \text{ and } n = 1. \end{aligned}$$

In the exceptional case, $K[\Lambda_1] = K$, and $\pi$ is certainly a norm. □

## Published mathematical erratum

*Milne's archived CFT errata, under "From Raghuram Sundararajan". The original proof above is reproduced as printed.*

(mathematical) In the proof of Theorem 3.6 (c) (page 39), that the uniformiser $\pi$ is a norm from $K_{\pi,n}$: "unless $q = 2$ and $n = 1$" should be "unless $q$ is even/(a power of 2) and $n = 1$". The argument given to cover that case then also needs to be rectified; this can be done by taking the norm of $-\pi_1$ rather than the norm of $\pi_1$. (More precisely, for $q$ a power of 2, $\mathrm{Nm}_{K_{\pi,n}/K_{\pi,n-1}}(\pm\pi_n) = \pi_{n-1}$ for $n \geq 2$, while $\mathrm{Nm}_{K_{\pi,1}/K}(\pm\pi_1) = \mp\pi$, so that $-\pi_n$ works for all $n$. In fact, $\mathrm{Nm}_{K_{\pi,n}/K}(-\pi_n) = \pi$ for any $q$ and $n$, so that the proof can be given without any separate edge cases.)

## Original supporting excerpts

*Source `CFT.pdf`, physical page 46, continuing from page 45.*
to take $f = \pi T + T^q$.[2] Then,

$$\begin{aligned}(f \circ f)(T) &\stackrel{\text{def}}{=} f(f(T)) = \pi(\pi T + \cdots + T^q) + \cdots + (\pi T + \cdots + T^q)^q \\ &= \pi^2 T + \cdots + T^{q^2},\end{aligned}$$

and

$$f^{(n)}(T) = \pi^n T + \cdots + T^{q^n}.$$

From the Newton polygon of $f^{(n)}$ (cf. ANT, 7.44), we see that the roots of $f^{(n)}$ all have positive $\text{ord}_K$, hence absolute value $< 1$, and so $\Lambda_n$ is the set of *all* roots of $f^{(n)}$ in $K^{\text{al}}$ endowed with the commutative group stucture

$$\alpha +_{F_f} \beta = F_f(\alpha, \beta) = \alpha + \beta + \cdots,$$

and the $A$-module structure,

$$[a]_f \alpha = a\alpha + \cdots.$$

EXAMPLE 3.2 Take $K = \mathbb{Q}_p$ and $f = (T+1)^p - 1 \in \mathcal{F}_p$; then

$$\Lambda_n = \{\alpha \in \mathbb{Q}_p^{\text{al}} \mid (\alpha - 1)^{p^n} = 1\} \simeq \{\zeta \mid \zeta^{p^n} = 1\} = \mu_{p^n}.$$

The addition $+_F$ on $\Lambda_n$ corresponds to multiplication on $\mu_{p^n}$, and the $\mathbb{Z}_p$-module structure is as defined before. It follows that $\Lambda_n \approx \mathbb{Z}/p^n\mathbb{Z}$ (as a $\mathbb{Z}_p$-module).

Recall that a ***cyclic module*** is a module that can be generated by a single element. Because $A$ is a principal ideal domain with only one prime element up to conjugates, every finitely generated torsion $A$-module $M$ decomposes into a direct sum of cyclic modules

$$M \approx A/(\pi^{n_1}) \oplus \cdots \oplus A/(\pi^{n_r}), \qquad n_1 \le n_2 \le \cdots,$$

and the sequence $n_1, \dots, n_r$ is uniquely determined.
LEMMA 3.3 *Let $M$ be an $A$-module, and let $M_n = \text{Ker}(\pi^n \colon M \longrightarrow M)$. Assume,*

*(a) $M_1$ has $q \stackrel{\text{def}}{=} (A : (\pi))$ elements, and*

*(b) $\pi \colon M \longrightarrow M$ is surjective.*

*Then $M_n \approx A/(\pi^n)$; in particular, it has $q^n$ elements.*

PROOF. We use induction on $n$. Because $A/\pi$ has order $q$, condition (a) and the structure theorem imply that $M_1 \approx A/(\pi)$. Consider the sequence

$$0 \longrightarrow M_1 \longrightarrow M_n \xrightarrow{\pi} M_{n-1} \longrightarrow 0.$$

Condition (b) implies that it is exact at $M_{n-1}$, and is therefore exact. It follows that $M_n$ has $q^n$ elements. If $M_n$ is not cyclic,

$$M_n \approx A/(\pi^{n_1}) \oplus A/(\pi^{n_2}) \oplus \cdots, \qquad n_1, n_2, \ldots \ge 1,$$

then $M_1$ is not cyclic,

$$M_1 \approx (\pi^{n_1 - 1})/(\pi^{n_1}) \oplus (\pi^{n_2 - 1})/(\pi^{n_2}) \oplus \cdots.$$

Therefore $M_n$ is a cyclic $A$-module of order $q^n$, and every such module is isomorphic to $A/(\pi^n)$. □

[2] Alternatively, the $p$-adic Weierstrass preparation theorem (Washington 1997, 7.3) implies that every $f \in \mathcal{F}_\pi$ factors into $f_1(T)u(T)$, where $f_1(T)$ is a polynomial $\pi T + \cdots + aT^q$, $a \equiv 1 \bmod \mathfrak{m}$, and $u(T)$ is a unit in $A[[T]]$. Obviously, the roots of $f^{(n)}$ are the roots of $f_1^{(n)}$.

*Source* **CFT.pdf**, *physical page 47.*

PROPOSITION 3.4 *The $A$-module $\Lambda_n$ is isomorphic to $A/(\pi^n)$. Hence* $\mathrm{End}_A(\Lambda_n) \simeq A/(\pi^n)$ *and* $\mathrm{Aut}_A(\Lambda_n) \simeq (A/(\pi^n))^\times$.

PROOF. An $A$-isomorphism $h\colon F_f \longrightarrow F_g$ of formal group laws induces an isomorphism of $A$-modules $\Lambda_f \longrightarrow \Lambda_g$, and so it does not matter which $f \in \mathcal{F}_\pi$ we choose. We take $f \in \mathcal{F}_\pi$ to be a polynomial of the form $\pi T + \cdots + T^q$. This is an Eisenstein polynomial (ANT, p. 129), and so has $q$ distinct roots, each with absolute value $< 1$. Let $\alpha \in K^{\mathrm{al}}$ have absolute value $< 1$. From the Newton polygon of

$$f(T) - \alpha = T^q + \cdots + \pi T - \alpha$$

we see that its roots have absolute value $< 1$, and so lie in $\Lambda_f$. We have shown that $\Lambda_f$ satisfies the hypotheses of the lemma, and so $\Lambda_n \approx A/(\pi^n)$. It follows that the action of $A$ on $\Lambda_n$ induces an isomorphism $A/(\pi^n) \longrightarrow \mathrm{End}_A(\Lambda_n)$. □

LEMMA 3.5 *Let $L$ be a finite Galois extension of a local field $K$, with Galois group $G$. For every $F \in \mathcal{O}_K[[X_1, \dots, X_n]]$ and $\alpha_1, \dots, \alpha_n \in \mathfrak{m}_L$,*

$$F(\tau\alpha_1, \dots, \tau\alpha_n) = \tau F(\alpha_1, \dots, \alpha_n), \qquad \text{all } \tau \in G.$$

PROOF. If $F$ is a polynomial, this follows from the fact that $\tau$ is a field isomorphism fixing the elements of $\mathcal{O}_K$. Because the absolute value on $L$ is the unique extension of the absolute value on $K$ (ANT, 7.38), it is preserved by $\tau \in G$, i.e., $|\tau\alpha| = |\alpha|$ all $\alpha \in L$, and so $\tau$ is continuous. Therefore it preserves limits,

$$\lim_{m\to\infty} \alpha_m = L \quad \Rightarrow \quad \lim_{m\to\infty} \tau\alpha_m = \tau L.$$

Write $F = F_m + \text{terms of degree } \geq m+1$ with $F_m$ a polynomial of degree $m$. Then

$$\begin{aligned} \tau(F(\alpha_1, \dots, \alpha_n)) &= \tau\big(\lim_{m\to\infty} F_m(\alpha_1, \dots, \alpha_n)\big) \\ &= \lim_{m\to\infty} \tau(F_m(\alpha_1, \dots, \alpha_n)) \\ &= \lim_{m\to\infty} F_m(\tau\alpha_1, \dots, \tau\alpha_n). \end{aligned}$$

□

*Published typographical erratum for the supporting Example 3.2. The archive attributes this entry to Raghuram Sundararajan.*

(typographical) In Example 3.2 (page 37), the first formula for $\Lambda_n$ should be $\{\alpha \mid (\alpha+1)^{p^n} = 1\}$, rather than $\{\alpha \mid (\alpha-1)^{p^n} = 1\}$.